LibreLift — Design Document

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Reviewers	[who reviewed it — a teammate, the instructor, a design-review partner]
Status	Draft
Last updated	10-16-26
Links	[proposal] · [repository] · [issue tracker]

Revision history

A design doc changes as the design does. Record each meaningful revision so a reader can tell what the doc said when a decision was made.

Date	Author	What changed
10-16-26	James Teuscher	First draft

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1. Context and scope

One or two paragraphs. What is being built, for whom, and what landscape it sits in. Link the proposal instead of repeating it — a reader who needs the full argument can follow the link.

2. Goals and non-goals

Goals — *what this design must achieve, stated so you could check them:*

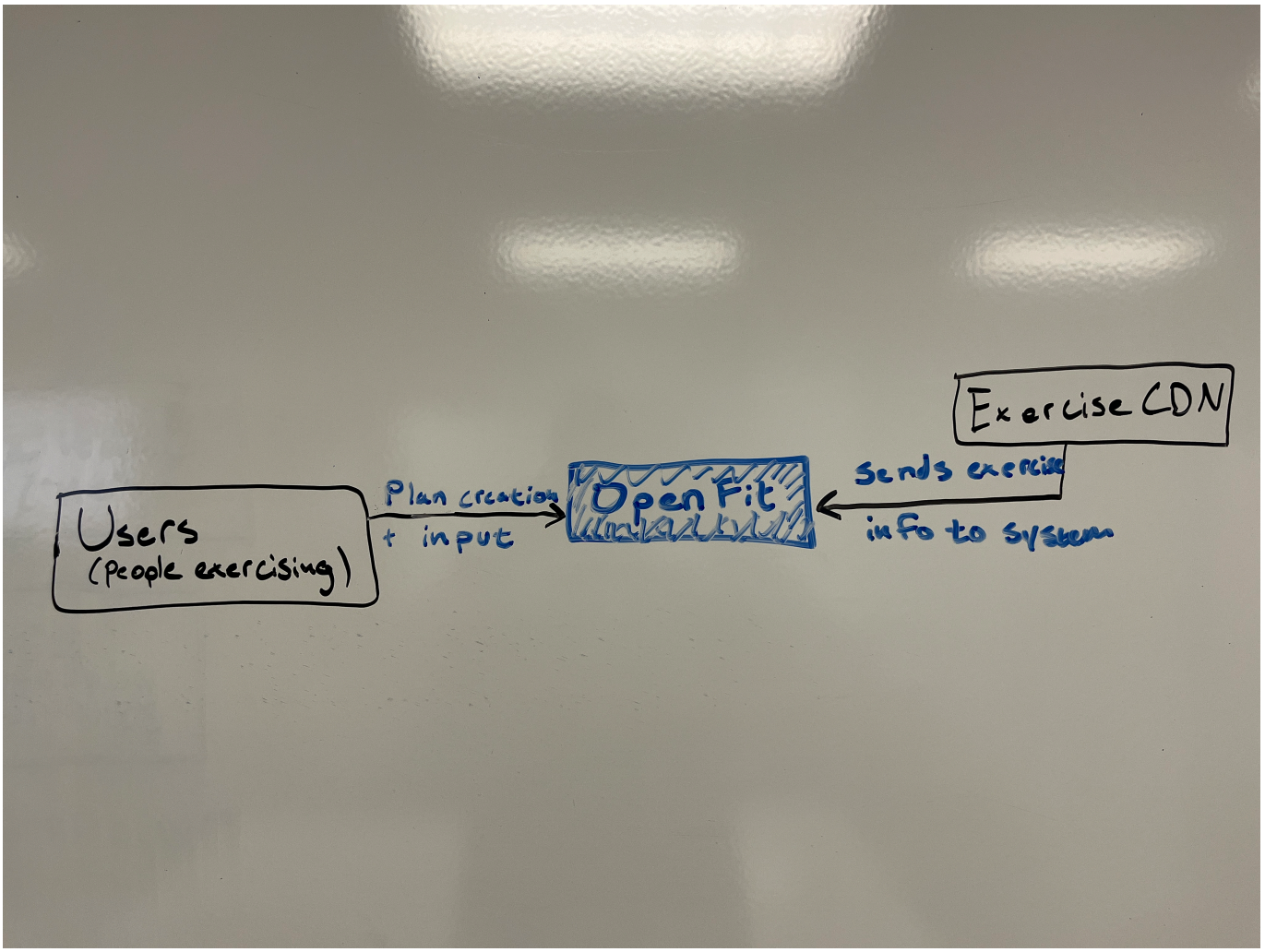
- [goal]

Non-goals — *things a reasonable reader would expect this system to do, which it deliberately won't. Carry them over from the proposal; here they bound the design.*

- [non-goal]

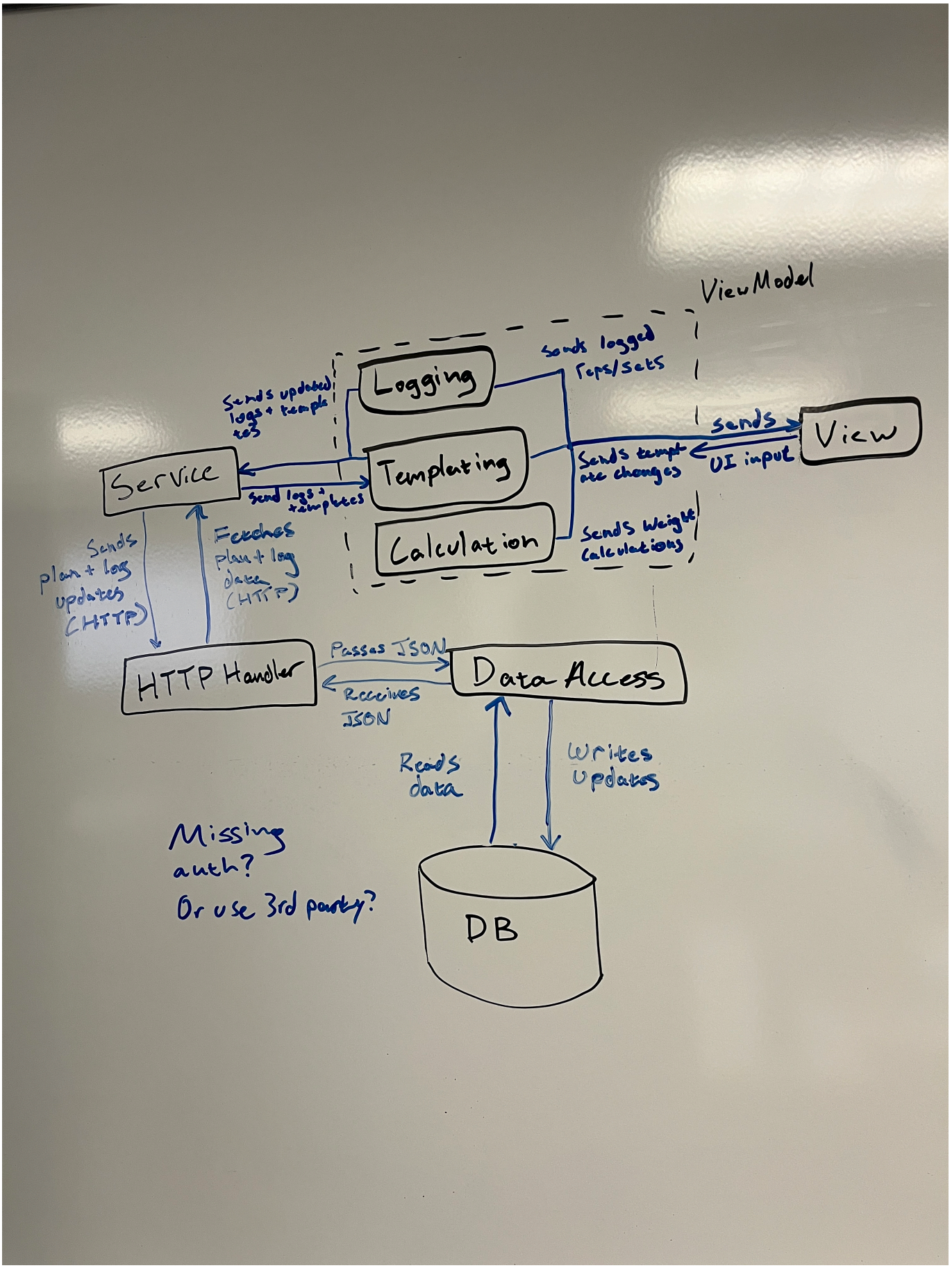
3. System-context diagram

There will only be one type of user interacting with this system. They will create plans and log workouts. The other external component interacting with this system is the CDN/API that provides a catalog of exercises. It will interact with the system by providing the exercise catalog via HTTP.



4. Architecture and components

Component	Responsibility	Talks to	Interface (rough)
View	Displays UI and accepts user input	ViewModel	function calls, touch screen elements
ViewModel	Business logic of logging, templating, and weight calculation	View, Service	function calls
Service	Makes calls to backend to read/write data for ViewModel operations	ViewModel, HTTPHandlers	function calls, HTTP
HTTPHandlers	Accepts incoming HTTP calls from frontend, handles request payloads	Service, DataAccess	function calls, HTTP
DataAccess	Adjusts HTTP payloads to be ready for DB reads/writes	DB, HTTPHandlers	function calls, HTTP
DB	Stores data persistently, supplies data for app UI	DataAccess	HTTP



Core flow, traced end to end

The single most important user action in your MVP, step by step through the components above. If the trace needs a component the table doesn't have, the table is wrong.

1. User taps the screen to create a plan
2. View layer sends input to ViewModel
3. ViewModel navigates to template creator business logic, calls service to load catalog
4. Service calls the Exercise CDN to load catalog, returns data to ViewModel
5. ViewModel sends exercises to View
6. View displays exercises
7. User chooses exercises, weight, reps
8. View sends selections to ViewModel
9. ViewModel saves plan in-memory as user builds template
10. User taps "Save plan"
11. View call ViewModel's save function
12. ViewModel sends plan data to service via function call
13. Service serializes the plan into JSON, calls backend via HTTP
14. HTTPHandler receives plan JSON, passes to DataAccess layer via function call
15. DataAccess layer deserializes JSON, writes to DB via HTTP

The failure case

The same flow when something goes wrong — the network drops, an external service is down, two people act at once. What does the user see, and where does the system recover?

1. User taps the screen to create a plan
2. View layer sends input to ViewModel
3. ViewModel navigates to template creator business logic, calls service to load catalog
4. Service checks if the catalog is cached. If not, call the Service layer
5. Service calls the Exercise CDN to load catalog, returns data to ViewModel
6. CDN throws an error and does not return catalog
7. Service passes an error to the ViewModel layer
8. ViewModel layer formats the error message to display on the UI
9. UI displays the error message for the User, asks to wait for a period and then try again

5. Data model

Entities, their key fields, and how they relate. Mark what the MVP needs and what only the roadmap needs. Key fields, not every column; no copied schemas.

Entity	Key fields	Relationships	MVP or roadmap
User	id	has Plans, performs Workouts	MVP
Workout	userId, workoutId?, exerciseIds?, title?	belongs to User, has exercises	MVP
Exercise	exerciseId, muscles, equipment	part of Workout	MVP

Note: One important distinction that I need to resolve is how to distinguish a template of a workout from a workout that is actively being done or is completed. My initial thought is a supertype of Workout and

Exercise, with subtypes/sub-entities, one for a template workout and one for a concrete, logged workout

6. Technology choices and alternatives considered

One block per significant decision. The alternative must be one a reasonable team would actually pick. Where you did real work to decide — a spike, a benchmark, a prototype — cite it; that evidence is the difference between a preference and a decision.

Decision: What ExerciseDB/CDN will be used

- **Chosen:** RepDB
- **Alternative considered:** yuhonas/free-exercise-db
- **Why the chosen option wins, given our goals:** There is a free starter license, and the example images are superior than those in free-exercise-db
- **Evidence:** The spike utilized RepDB, while a personal side project has utilized free-exercise-db, and the RepDB images are superior, while muscle grouping, exercises available, and instructions are roughly equivalent.
- **What would change our mind:** If the user base grows enough that the RepDB free license is no longer usable, I would want to switch to the free alternative of free-exercise-db

7. Risks and open questions

What you don't know yet, stated honestly. Each open question gets an owner and a way to find the answer.

Risk or question	Why it matters	Owner	How we'll find out, and by when
[risk]	[consequence]	[name]	[spike / test / conversation — milestone]

8. Implementation plan

The build order at milestone altitude — not a task list. Start with the thinnest vertical slice through the core flow, then widen it.

Milestone	What works when it's done	Depends on
1	[the thinnest end-to-end slice]	—

Sign-off

Who reviewed this version and agreed it is ready to build from.

Reviewer	Role	Date	Approved
[name]	[role]	[date]	□